

- 1 (a) Use of $mgh = \frac{1}{2} m v^2$ **and** makes h subject C1
 $h = 14.7$ or 15 (m) A1
- (b) (i) Calculate the final vertical velocity at C (using $v = 0 + at = 9.81 \times 1.6$) C1
 $v = 15.7$ or 16 (m s^{-1}) A1
- (ii) Straight line of positive gradient) $\square$ A1
 Starting at 0 ms^{-1} and ends at 16 ms^{-1} at 1.6 s . A1
- (b) (iii) Use of pythagoras' theorem: M1
 $\text{resultant } v^2 = 15.7^2 + 17^2$
 $v = 23$ or 23.1 m s^{-1}
- (c) slope: smaller change in vertical component of velocity/ B1
 smaller change in vertical component of momentum
 by Newton's second law, the force experienced = rate of change B1
 of momentum is less, so less risk of injury
- Bonus mark: if she bends her knees during landing, she
 increases time for (same) change of momentum, so force
 exerted on her is even lesser.